

6

The Fast Fourier Transform

DFT is Slow due to Complex Computations

- From

$$X(k) = \sum_{n=0}^{N-1} x(n)W_N^{kn}, \quad k = 0, 1, \dots, N-1$$

- Where

$$W_N \triangleq e^{-j\frac{2\pi}{N}}$$

- This is the source of the problem since the W_N is a complex number; it needs too many multiplies and additions.

Example 6.1: Determine Numbers of Multiplications and Additions

$$XY \quad \begin{aligned} X &= a_1 + jb_1 \\ Y &= a_2 + jb_2 \end{aligned}$$

$$(a_1 + jb_1).(a_2 + jb_2) = c_1 + jd_1$$

Four Multiplications

$$(a_1 + jb_1).(a_2 + jb_2) = a_1a_2 - b_1b_2 + j(b_1a_2 + a_1b_2)$$

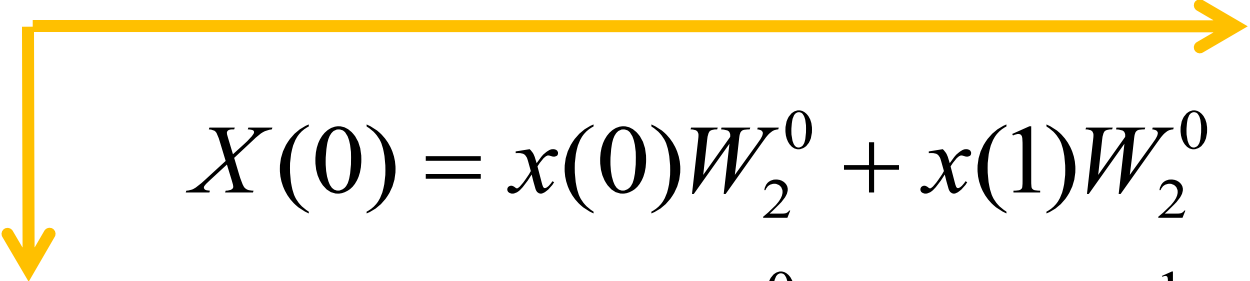
The diagram illustrates the expansion of the complex multiplication formula. A central point has four red arrows pointing to the terms a_1a_2 , $-b_1b_2$, $j(b_1a_2 + a_1b_2)$, and the closing parenthesis. Below this, a point has three red arrows pointing to the terms a_1a_2 , $-b_1b_2$, and $j(b_1a_2 + a_1b_2)$, representing the three additions required to sum these products.

Three Additions

Example 6.2 For a 2-point DFT

- Solution:
$$\begin{bmatrix} X(0) \\ X(1) \end{bmatrix} = \begin{bmatrix} W_2^0 & W_2^0 \\ W_2^0 & W_2^1 \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \end{bmatrix}$$

2 Multiplies per each k


$$\begin{aligned} X(0) &= x(0)W_2^0 + x(1)W_2^0 \\ X(1) &= x(0)W_2^0 + x(1)W_2^1 \end{aligned}$$

k = 2

Example 6.2: Find number of multiplications for a 4-point DFT

• Solution:
$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

4 multiplies

$$X(0) = x(0)W_4^0 + x(1)W_4^0 + x(2)W_4^0 + x(3)W_4^0$$

$$X(1) = x(0)W_4^0 + x(1)W_4^1 + x(2)W_4^2 + x(3)W_4^3$$

$$X(2) = x(0)W_4^0 + x(1)W_4^2 + x(2)W_4^4 + x(3)W_4^6$$

$$X(3) = x(0)W_4^0 + x(1)W_4^3 + x(2)W_4^6 + x(3)W_4^9$$

4 Times

A Simple way to reduce the multiplications

- For example, for a 2 point DFT (N=2)

$$N = 2 \quad \rightarrow \quad W_2 = e^{-j\pi}$$

$$X(k) = \sum_{n=0}^1 x(n)W_2^{kn} = x(0)W_2^{0.k} + x(1)W_2^{1.k}, \quad k = 0, 1$$

$$X(0) = x(0).W_2^{0.0} + x(1)W_2^{1.0}$$

$$X(1) = x(0).W_2^{1.0} + x(1)W_2^{1.1}$$

A simple way to reduce the multiplications

- However, if we pre-calculate W_2 as

$$N = 2 \quad \rightarrow \quad W_2 = e^{-j\pi} = -1$$

$$X(k) = \sum_{n=0}^1 x(n)W_2^{kn} = x(0)W_2^{0.k} + x(1)W_2^{1.k}, \quad k = 0, 1$$

$$X(0) = x(0) \cdot 1 + x(1) \cdot 1$$

$$X(1) = x(0) \cdot 1 + x(1) \cdot (-1)$$

Pre-calculated 2-point DFT

- So the 2-point DFT has no complex multiplies with

$$X(0) = x(0) + x(1)$$

$$X(1) = x(0) - x(1)$$

- But this is not the case when N is not two.

The Fast Fourier Transform (FFT)

- FFT is a collective name for any algorithm for fast DFT computations.
- The Divide and conquer approach is to be used.
- We consider the method of **Decimation in time or DIT.**

Radix-2

- Radix-2 DIT divides a DFT of size N into two interleaved DFTs of size $N/2$.
- It works when N is of "base two" or Radix-2.
- Therefore it is called **Radix-2 DIT-FFT**.

$x(n)$ for 4-point DFT ($N=4$)

- For $x(n) = \{x(0), x(1), x(2), x(3)\}$

$x(0)$	$x(1)$	$x(2)$	$x(3)$
--------	--------	--------	--------

- Reorder $x(n)$ to even and odd groups:

$x(0)$	$x(2)$	$x(1)$	$x(3)$
--------	--------	--------	--------

$x()$

= Even indices

$x()$

= Odd indices

4-point DFT

- For a reordered

$$X(k) = \sum_{n=0}^3 x(n) W_4^{kn}, \quad k = 0, 1, 2, 3$$

$$= x(0) W_4^{k \cdot 0} + x(1) W_4^{k \cdot 1} + x(2) W_4^{k \cdot 2} + x(3) W_4^{k \cdot 3}$$

$$= x(0) W_4^{k \cdot 0} + x(2) W_4^{k \cdot 2} + x(1) W_4^{k \cdot 1} + x(3) W_4^{k \cdot 3}$$

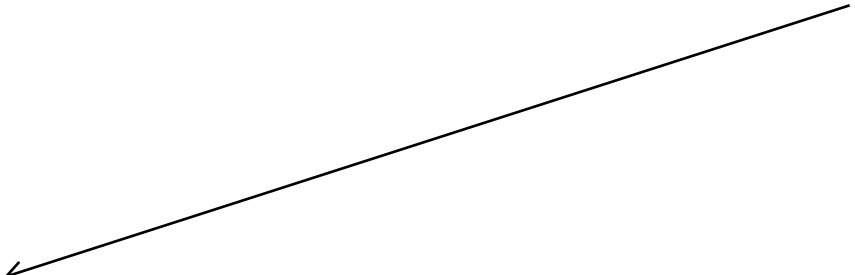
Grouping Terms

$$X(k) = \sum_{n=0}^3 x(n)W_4^{kn}, \quad k = 0, 1, 3, 4$$

$$= x(0)W_4^{k.0} + x(2)W_4^{k.2} + x(1)W_4^{k.1} + x(3)W_4^{k.3}$$

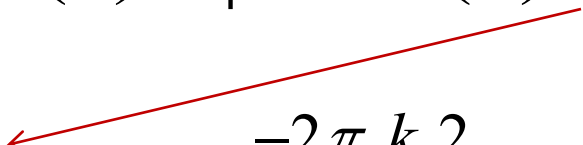
$$= \sum_{n=0}^1 x(2n)W_4^{kn} + \sum_{n=0}^1 x(2n+1)W_4^{kn}$$

$$X(k) = x(0)W_4^{k.0} + x(2)W_4^{k.2} + x(1)W_4^{k.1} + x(3)W_4^{k.3}$$



$$W_4^{k.3} = W_4^{k.(2+1)} = W_4^k W_4^{k.2}$$

$$X(k) = x(0)W_4^{k.0} + x(2)W_4^{k.2} + W_4^{k.1}[x(1)W_4^{k.0} + x(3)W_4^{k.2}]$$



$$W_4^{k.2} = e^{\frac{-2\pi.k.2}{4}} = e^{\frac{-2\pi.k}{4/2}} = e^{\frac{-2\pi.k}{2}} = W_2^k$$

$$W_N^{k.2} = W_{N/2}^k$$

$$X(k) = \underbrace{x(0)W_2^{k.0} + x(2)W_2^{k.1}}_{\text{Yellow}} + W_4^{k.1} \underbrace{[x(1)W_2^{k.0} + x(3)W_2^{k.1}]}_{\text{Blue}}$$

$$X_1(k) = \sum_{n=0}^1 x(2n)W_2^{kn}, \quad k = 0, 1, 2, 3$$

$$X_2(k) = \sum_{n=0}^1 x(2n+1)W_2^{kn}, \quad k = 0, 1, 2, 3 \quad \leftarrow$$

$$X(0) = \sum_{n=0}^1 x(2n)W_2^{0 \cdot n} + W_4^{0.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{0 \cdot n} \right]$$

$$X(1) = \sum_{n=0}^1 x(2n)W_2^{1 \cdot n} + W_4^{1.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{1 \cdot n} \right]$$

$$X(2) = \sum_{n=0}^1 x(2n)W_2^{2 \cdot n} + W_4^{2.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{2 \cdot n} \right]$$

$$X(3) = \sum_{n=0}^1 x(2n)W_2^{3 \cdot n} + W_4^{3.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{3 \cdot n} \right]$$

The 4-point DFT $X(k)$, $k=0,1,2,3$

$$X(0) = X_1(0) + W_4^0 X_2(0)$$

$$X(1) = X_1(1) + W_4^1 X_2(1)$$

$$X(2) = X_1(2) + W_4^2 X_2(2)$$

$$X(3) = X_1(3) + W_4^3 X_2(3)$$

Reduced Further using Periodicity of W_N

$$W_2^{2.n} = e^{\frac{-2\pi.(0+2).n}{2}} = e^{\frac{-2\pi.0.n}{2}} \underbrace{e^{\frac{-2\pi.2.n}{2}}}_{=1} = W_2^0$$

$$W_2^{3.n} = e^{\frac{-2\pi.(1+2).n}{2}} = e^{\frac{-2\pi.1.n}{2}} \underbrace{e^{\frac{-2\pi.2.n}{2}}}_{=1} = W_2^1$$

Therefore,

$$X(0) = \sum_{n=0}^1 x(2n)W_2^{0 \cdot n} + W_4^{0.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{0 \cdot n} \right]$$

$$X(1) = \sum_{n=0}^1 x(2n)W_2^{1 \cdot n} + W_4^{1.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{1 \cdot n} \right]$$

$$X(2) = \sum_{n=0}^1 x(2n)W_2^{0 \cdot n} + W_4^{2.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{0 \cdot n} \right]$$

$$X(3) = \sum_{n=0}^1 x(2n)W_2^{1 \cdot n} + W_4^{3.1} \left[\sum_{n=0}^1 x(2n+1)W_2^{1 \cdot n} \right]$$

The 4-point DFT $X(k)$, $k=0,1,2,3$

$$X(0) = X_1(0) + W_4^0 X_2(0)$$

$$X(1) = X_1(1) + W_4^1 X_2(1)$$

$$X(2) = X_1(0) + W_4^2 X_2(0)$$

$$X(3) = X_1(1) + W_4^3 X_2(1)$$

General

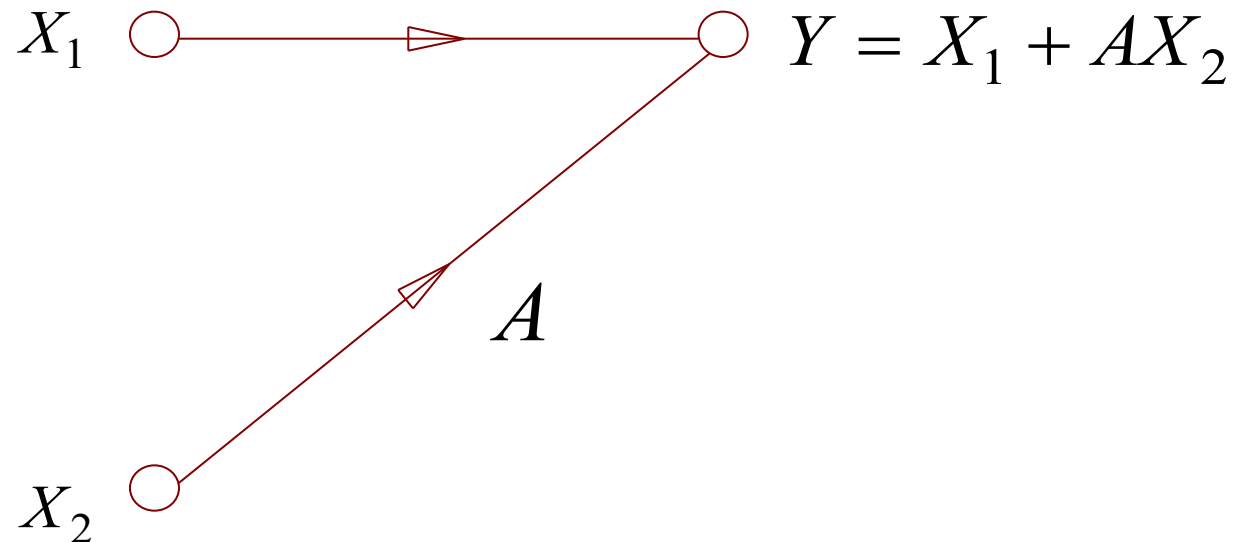
$$W_{N/2}^{(k+N/2).n} = e^{\frac{-2\pi.(k+N/2).n}{N/2}} = e^{\frac{-2\pi.k.n}{N/2}} \underbrace{e^{\frac{-2\pi.(N/2).n}{N/2}}}_{=W_{N/2}^{0.n}=1} = W_{N/2}^k$$

The Cooley–Tukey algorithm

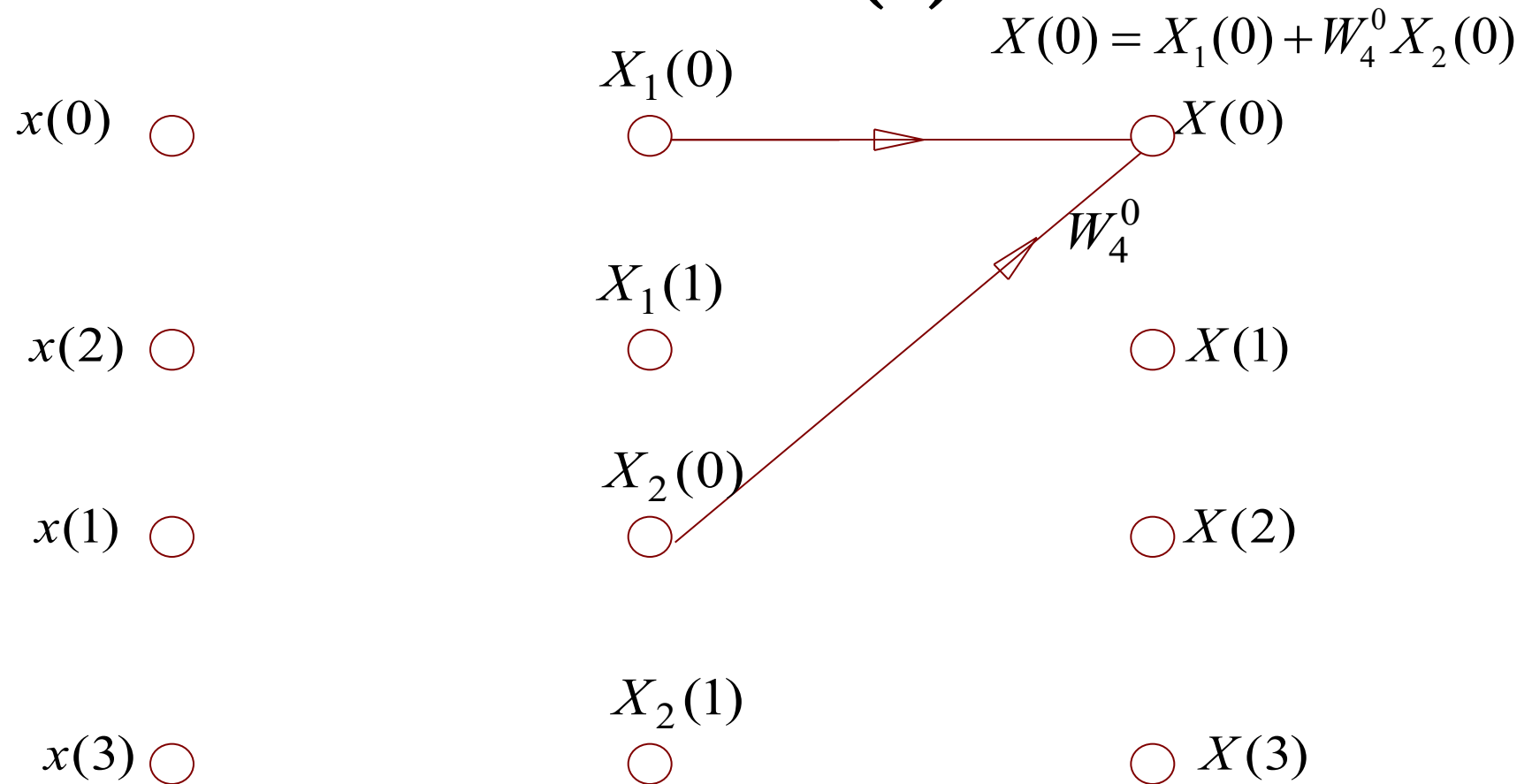
- It breaks the DFT into smaller DFTs.
- (From Wikipedia) Tukey reportedly came up with the idea during a meeting of President Kennedy's Science Advisory Committee discussing ways to detect nuclear-weapon tests in the Soviet Union by employing seismometers located outside the country.
- However, analysis of this data would require fast algorithms for computing DFT due to number of sensors and length of time: *Hence FFT was born.*

Signal Flow Graph

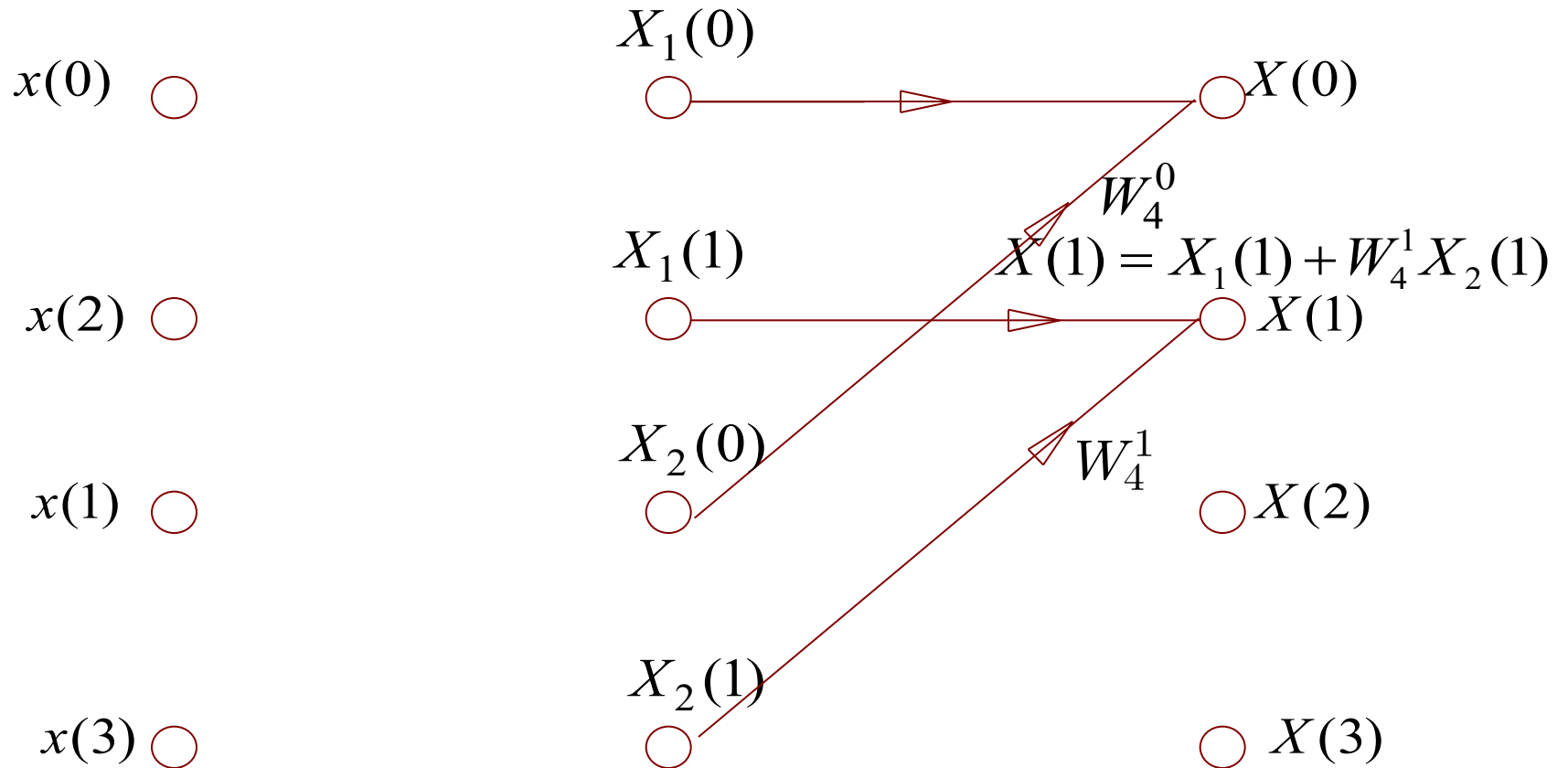
- The FFT algorithm is best represented by the signal flow graph.



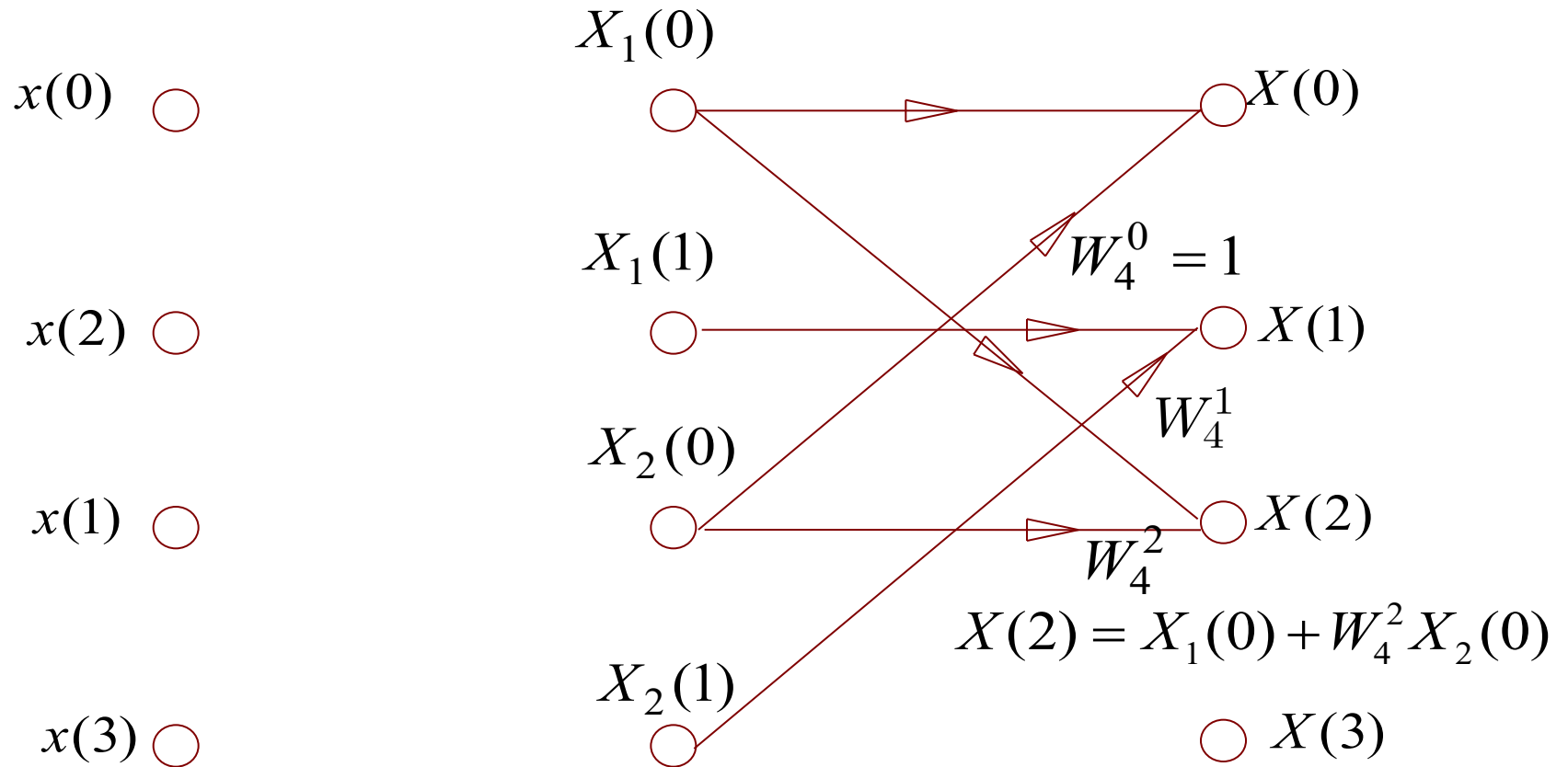
Determine $X(0)$



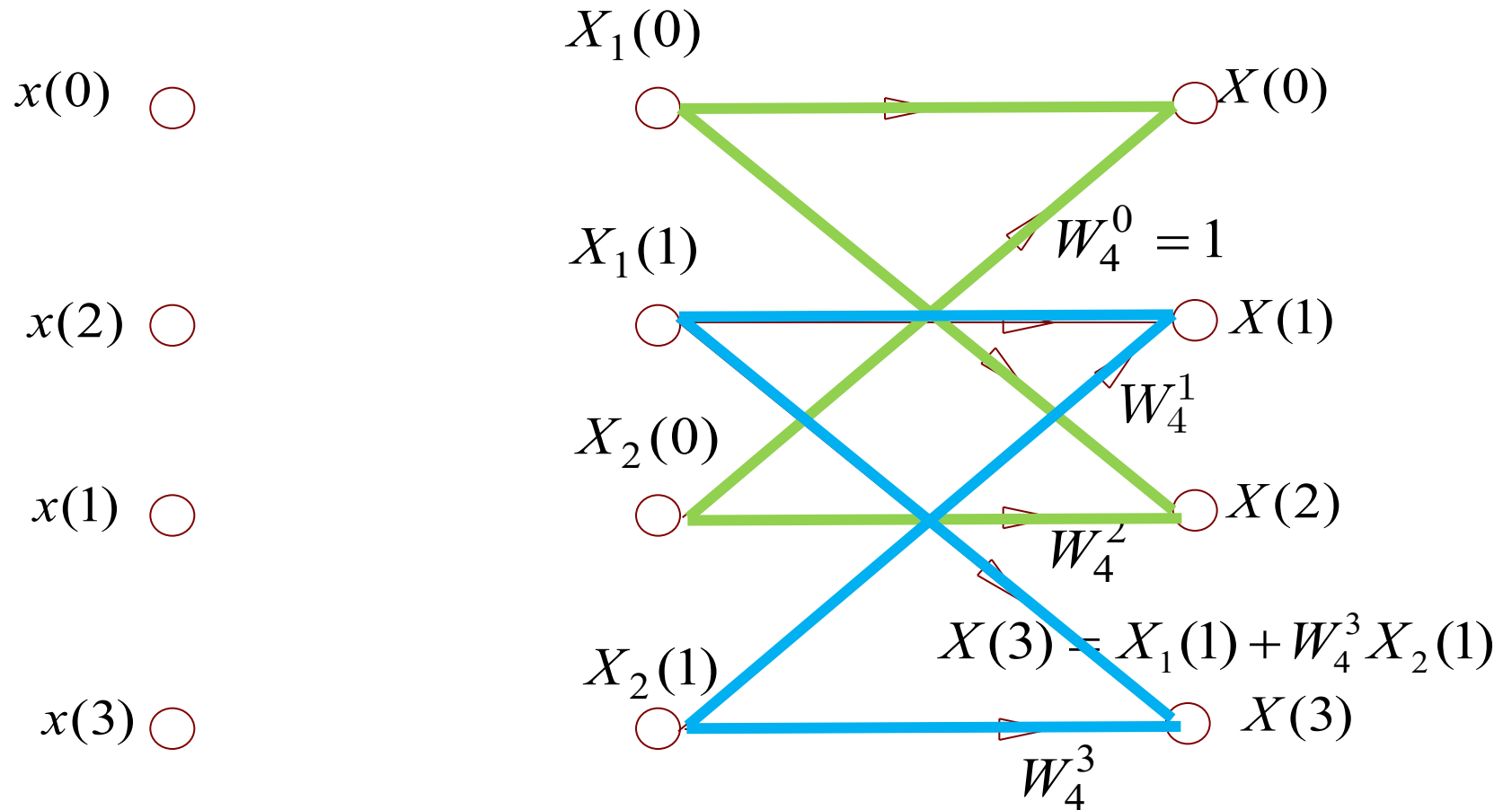
Determine $X(1)$

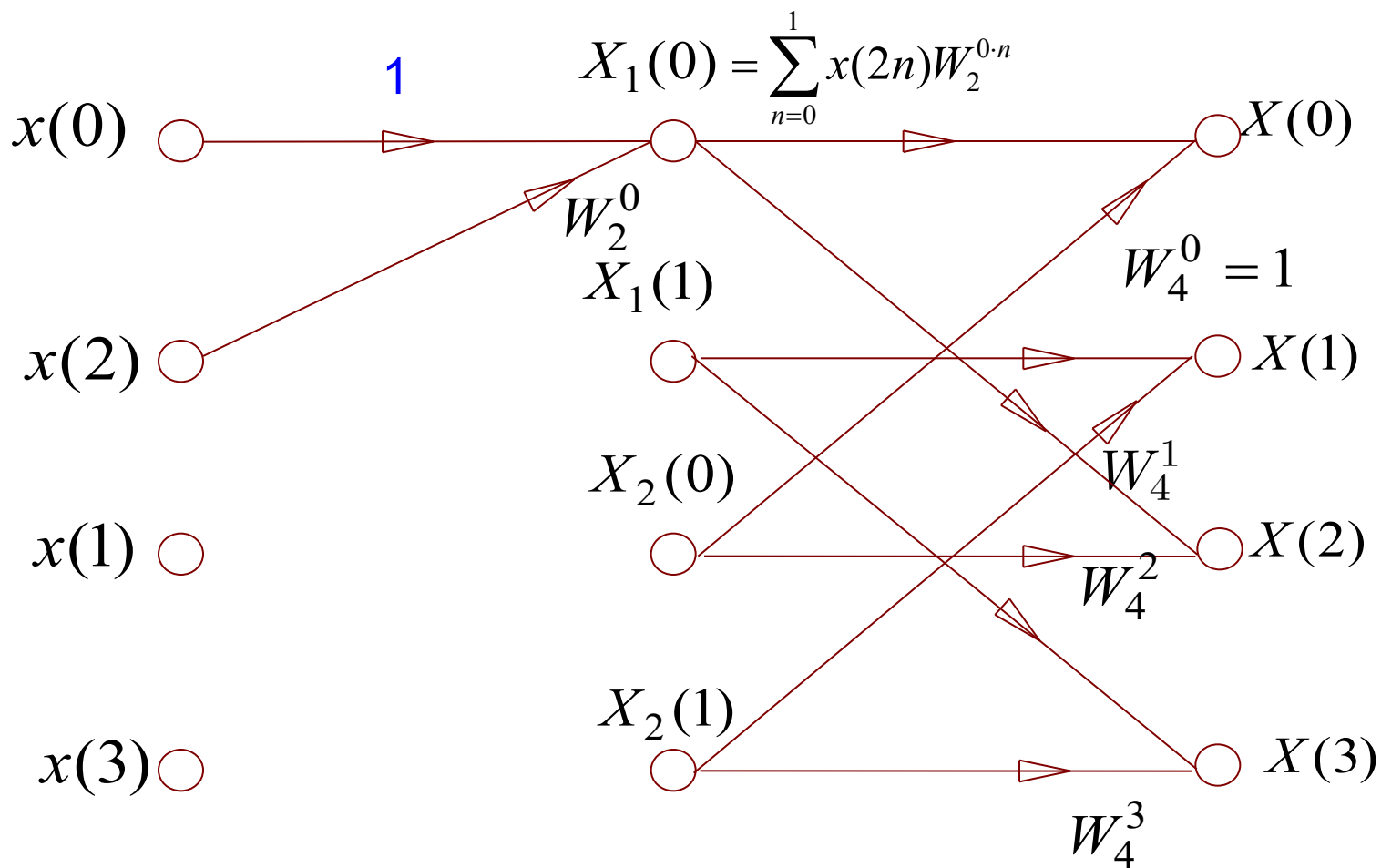


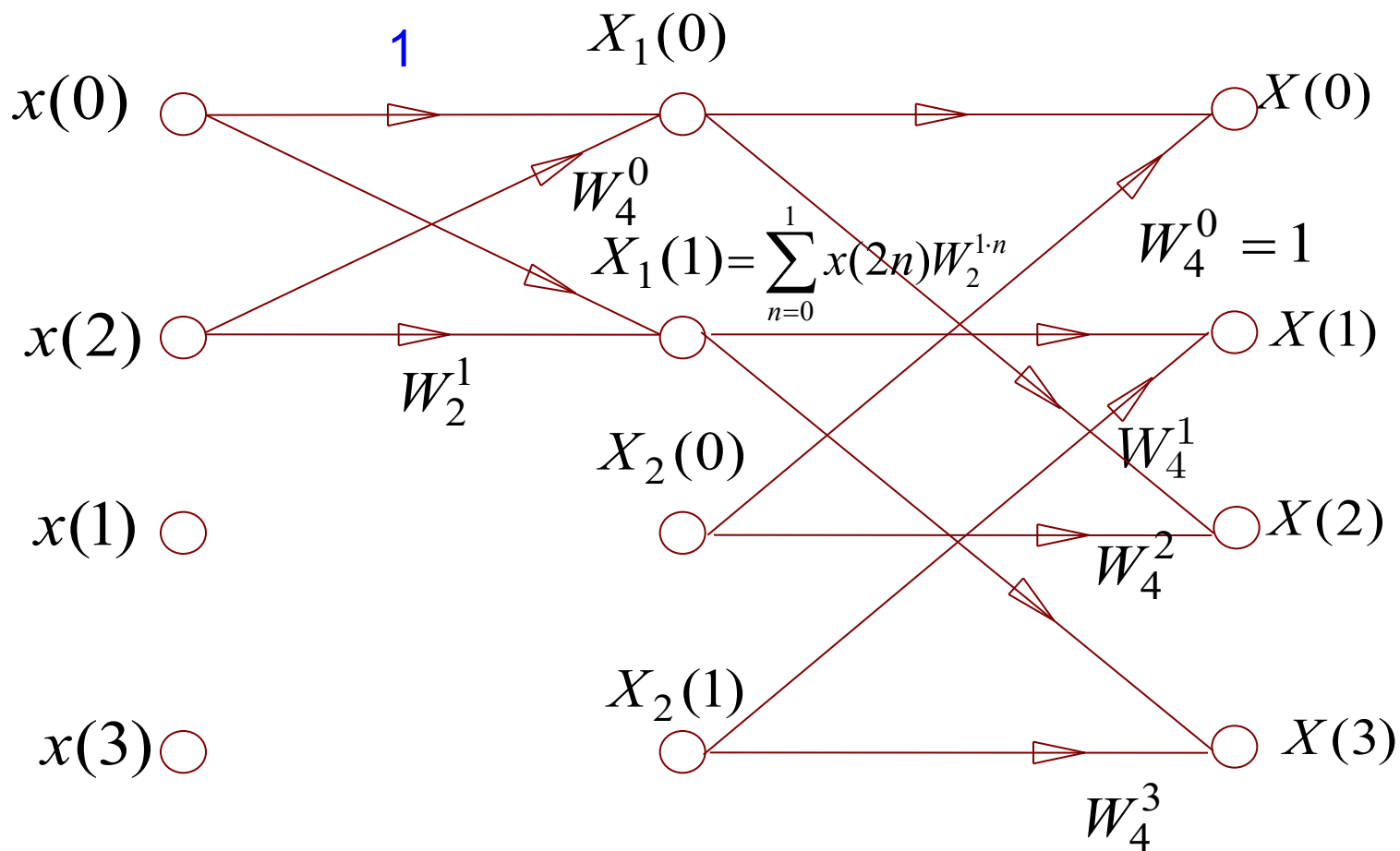
Determine $X(2)$



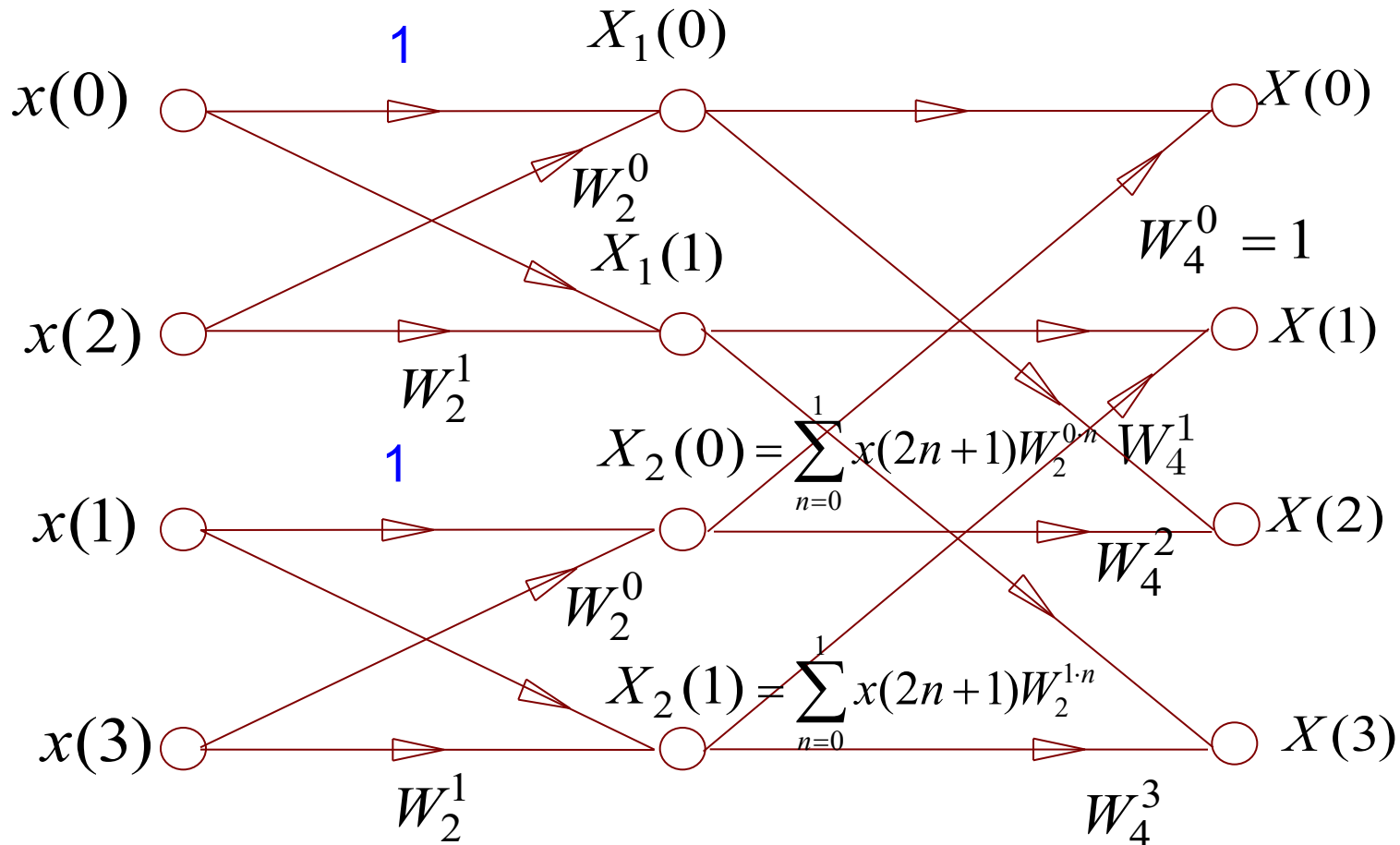
Determine $X(3)$







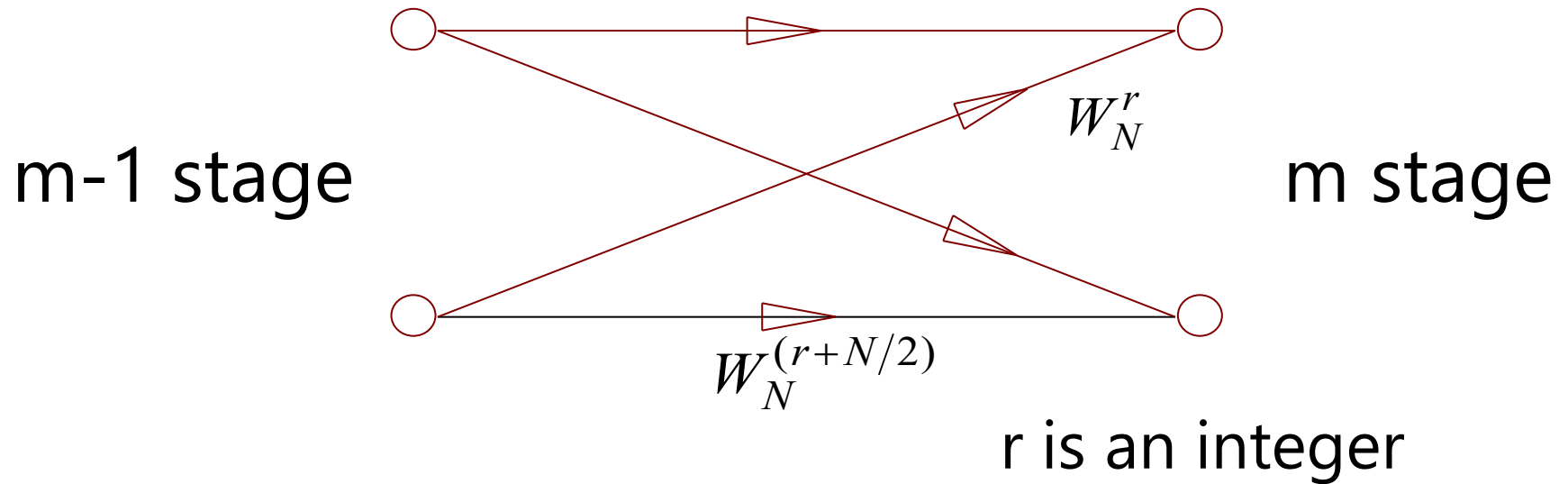
The 4-point DIT-FFT



2-point DFT x 2

The Butterfly

- It is basically a signal flow graph.
- Two complex multiplies per butterfly.



$x(n)$ for 8-point DFT ($N=8$)

$x(0)$	$x(1)$	$x(2)$	$x(3)$	$x(4)$	$x(5)$	$x(6)$	$x(7)$
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Reordering

$x(0)$	$x(2)$	$x(4)$	$x(6)$	$x(1)$	$x(3)$	$x(5)$	$x(7)$
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8-point DIT-FFT

$$N = 8 \rightarrow W_8 = e^{-j\frac{2\pi}{8}} = e^{-j\frac{\pi}{4}} = \sqrt{2} - j\sqrt{2}$$

$$\begin{aligned} X(k) &= \sum_{n=0}^7 x(n)W_8^{nk}, \quad k = 0, \dots, 7 \\ &= x(0)W_8^{0.k} + x(1)W_8^{1.k} + x(2)W_8^{2.k} + x(3)W_8^{3.k} \\ &\quad + x(4)W_8^{4.k} + x(5)W_8^{5.k} + x(6)W_8^{6.k} + x(7)W_8^{7.k} \end{aligned}$$

Reordering

$$\begin{aligned} X(k) &= x(0)W_8^{0.k} + x(2)W_8^{2.k} + x(4)W_8^{4.k} + x(6)W_8^{6.k} \\ &\quad + x(1)W_8^{1.k} + x(3)W_8^{3.k} + x(5)W_8^{5.k} + x(7)W_8^{7.k} \end{aligned}$$

8-point DIT-FFT

$$X(k) = x(0)W_8^{k.0} + x(2)W_8^{k.2} + x(4)W_8^{k.4} + x(6)W_8^{k.6} \\ + W_8^{k.1} \left[x(1)W_8^{k.0} + x(3)W_8^{k.2} + x(5)W_8^{k.4} + x(7)W_8^{k.6} \right]$$

$$W_N^2 = e^{j\frac{2\pi.2}{N}} = e^{j\frac{2\pi}{\frac{N}{2}}} = W_{N/2}$$

$$X(k) = \underbrace{x(0)W_4^{k.0} + x(2)W_4^{k.1} + x(4)W_4^{k.2} + x(6)W_4^{k.3}}_{X_1(k)} \\ + W_8^{k.1} \left[\underbrace{x(1)W_4^{k.0} + x(3)W_4^{k.1} + x(5)W_4^{k.2} + x(7)W_4^{k.3}}_{X_2(k)} \right]$$

8-point DIT-FFT

- For $k=0:8$

$$X(0) = X_1(0) + W_8^0 X_2(0)$$

$$X(4) = X_1(4) + W_8^4 X_2(4)$$

$$X(1) = X_1(1) + W_8^1 X_2(1)$$

$$X(5) = X_1(5) + W_8^5 X_2(5)$$

$$X(2) = X_1(2) + W_8^2 X_2(2)$$

$$X(6) = X_1(6) + W_8^6 X_2(6)$$

$$X(3) = X_1(3) + W_8^3 X_2(3)$$

$$X(7) = X_1(7) + W_8^7 X_2(7)$$

Reduced form of $X_1(k+4)$

$$\begin{aligned} X_1(k+4) &= x(0)W_4^{(k+4).0} + x(2)W_4^{(k+4).1} + x(4)W_4^{(k+4).2} + x(6)W_4^{(k+4).3} \\ &= x(0)W_4^{k.0}W_4^{4.0} + x(2)W_4^{k.1}W_4^{4.1} + x(4)W_4^{k.2}W_4^{4.2} + x(6)W_4^{k.3}W_4^{4.3} \end{aligned}$$

$$W_4^{4.n} = e^{-j\frac{2\pi}{4}(4 \cdot n)} = (e^{-j2\pi})^n = (1)^n = 1$$

$$\begin{aligned} &= x(0)W_4^{k.0} + x(2)W_4^{k.1} + x(4)W_4^{k.2} + x(6)W_4^{k.3} \\ &= X_1(k) \end{aligned}$$

$$\therefore X_1(k+4) = X_1(k)$$

The same reduction is true for $X_2(k+4) = X_2(k)$.

Further reduction

- We know that

$$X_1(k) = X_1(k+4) \quad X_2(k) = X_2(k+4)$$

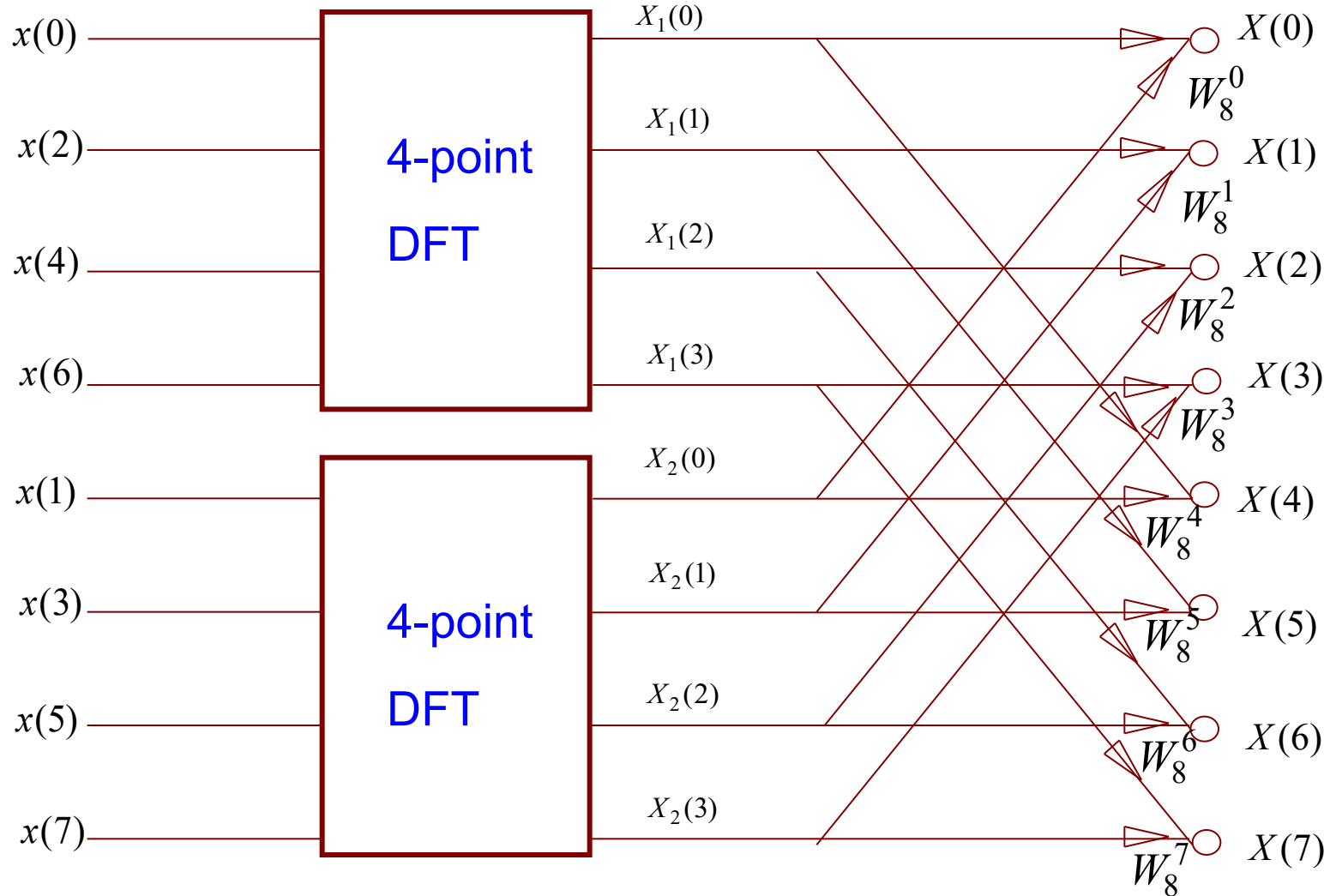
$$X(0) = X_1(0) + W_8^0 X_2(0) \quad X(4) = X_1(0) + W_8^4 X_2(0)$$

$$X(1) = X_1(1) + W_8^1 X_2(1) \quad X(5) = X_1(1) + W_8^5 X_2(1)$$

$$X(2) = X_1(2) + W_8^2 X_2(2) \quad X(6) = X_1(2) + W_8^6 X_2(2)$$

$$X(3) = X_1(3) + W_8^3 X_2(3) \quad X(7) = X_1(3) + W_8^7 X_2(3)$$

8-point



Further Reduction

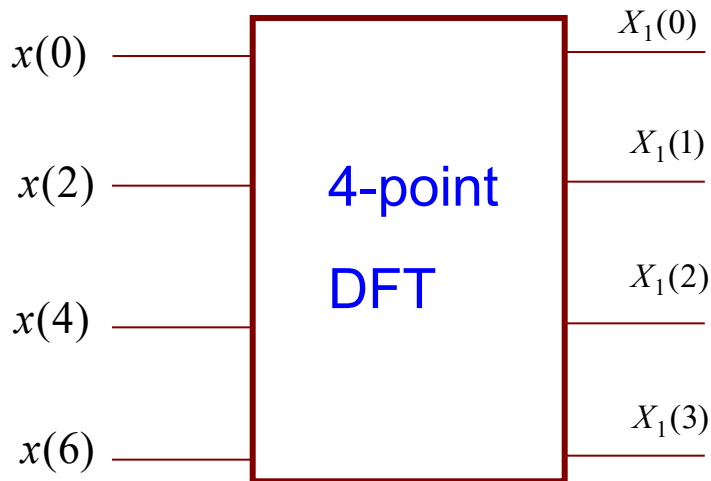
$$X(k) = \boxed{X_1(k)} - W_8^{k.1} X_2(k), \quad k = 0, \dots, 7$$

$$X_1(k) = x(0)W_4^{k.0} + x(2)W_4^{k.1} + x(4)W_4^{k.2} + x(6)W_4^{k.3}$$

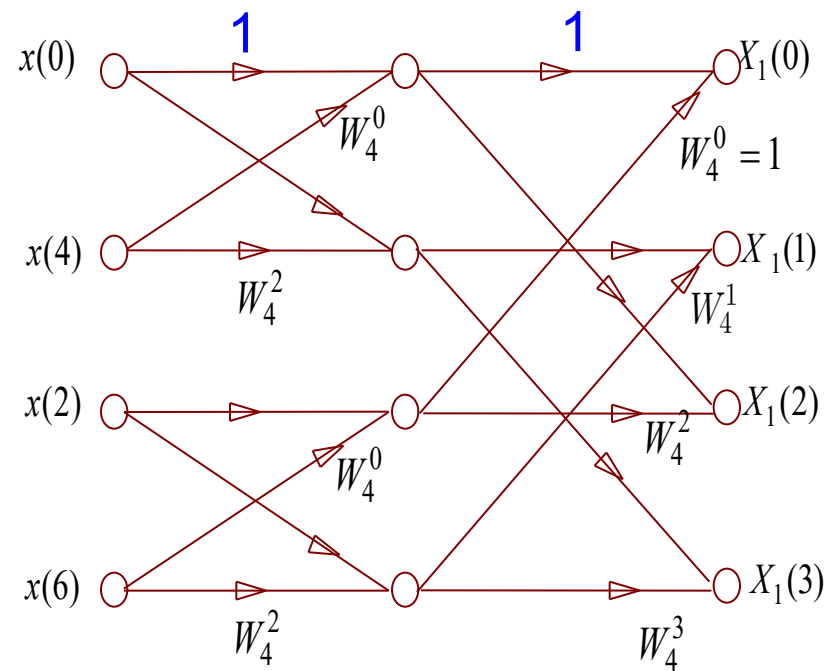
$$= x(0)W_4^{k.0} + x(2)W_4^{k.1} + x(4)W_4^{k.2} + x(6)W_4^{k.3}$$

$$= x(0)W_4^{k.0} + x(4)W_4^{k.2} + W_4^{k.1} \left[x(2)W_4^{0.k} + x(6)W_4^{k.2} \right]$$

2-point DFT $x(0), x(2), x(4)$ and $x(6)$



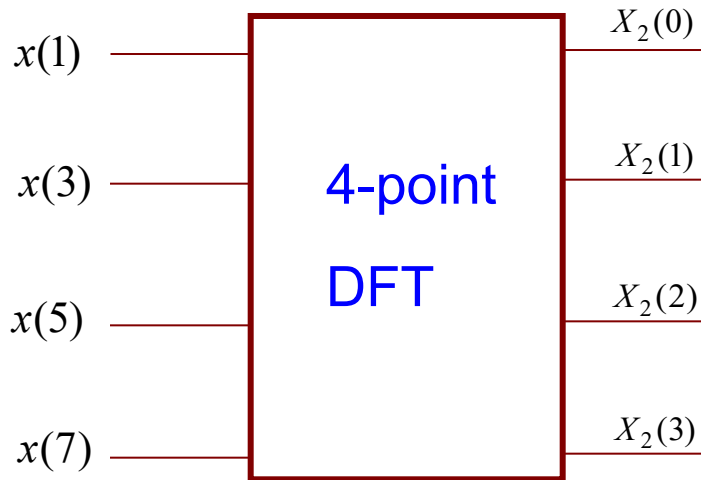
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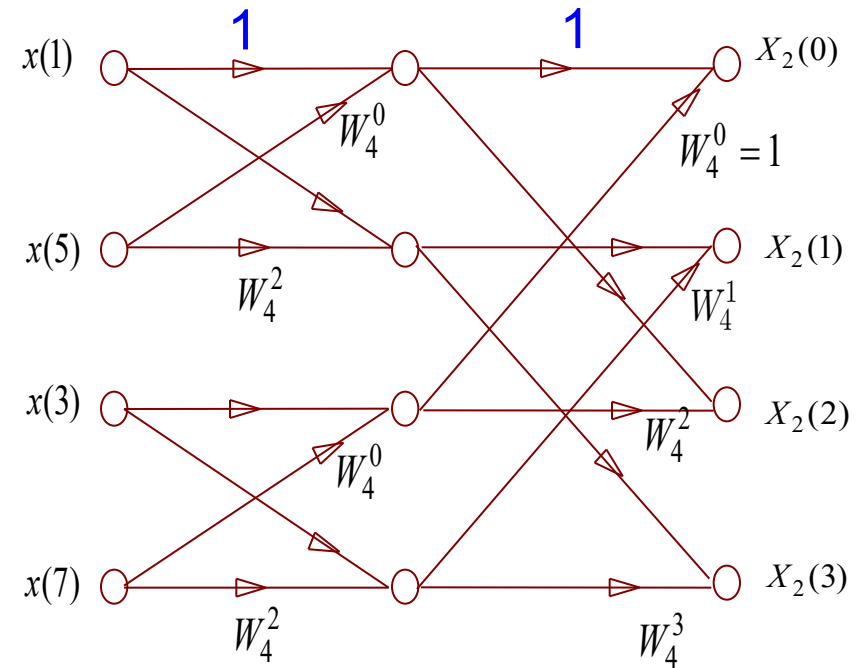
$$X(k) = X_1(k) + W_8^{k.1} X_2(k), \quad k = 0, \dots, 7$$

$$\begin{aligned} X_2(k) &= x(1)W_4^{k.0} + x(3)W_4^{k.1} + x(5)W_4^{k.2} + x(7)W_4^{k.3} \\ &= x(1)W_4^{0.k} + x(5)W_4^{k.2} + x(3)W_4^{k.1} + x(7)W_4^{k.3} \\ &= x(1)W_4^{0.k} + x(5)W_4^{k.2} + W_4^{k.1} \left[x(3)W_4^{k.1} + x(7)W_4^{k.3} \right] \end{aligned}$$

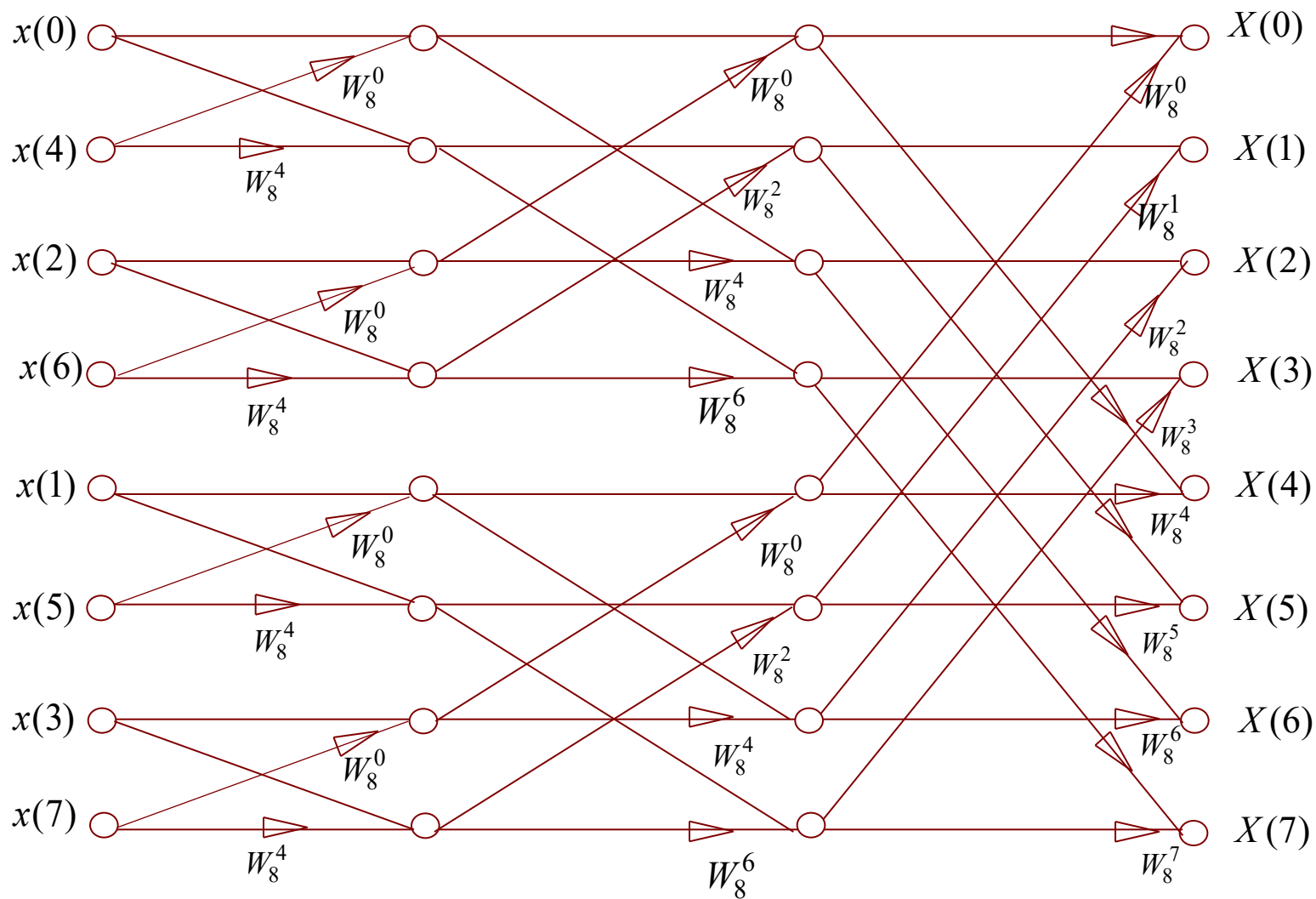
2-point DFT $x(1), x(3), x(5)$ and $x(7)$



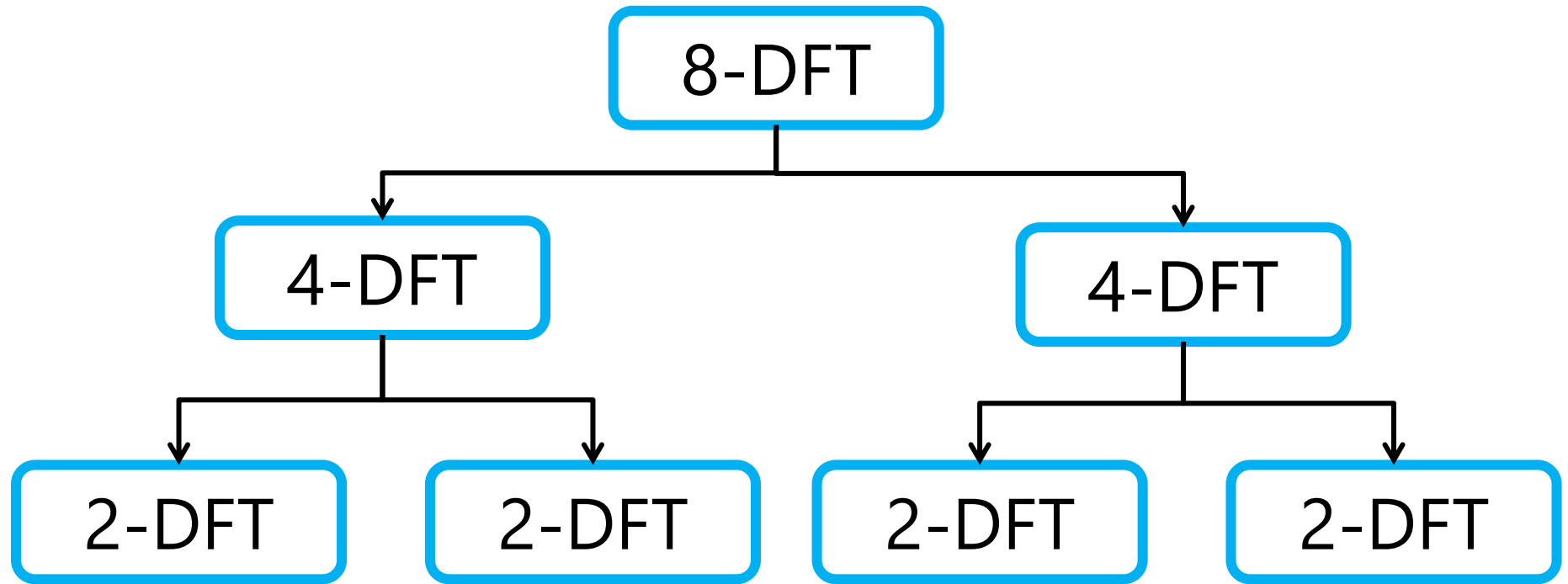
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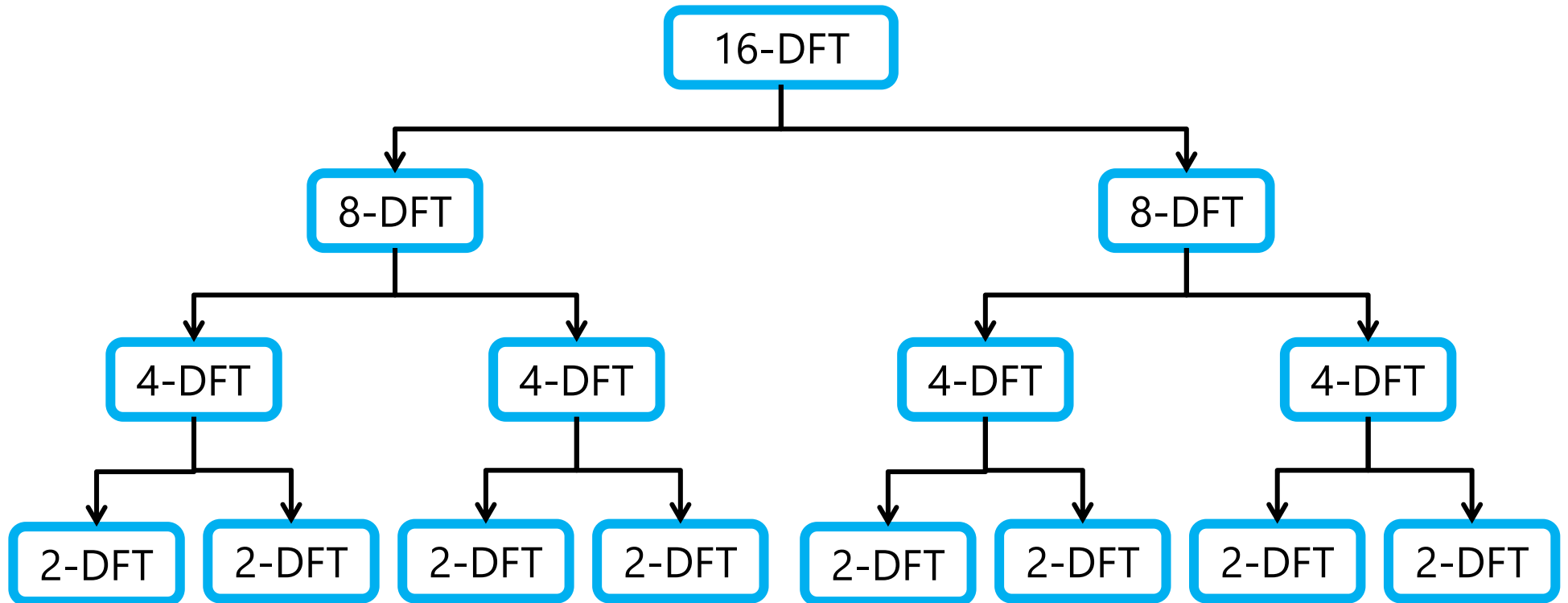
8-point DIT-FFT



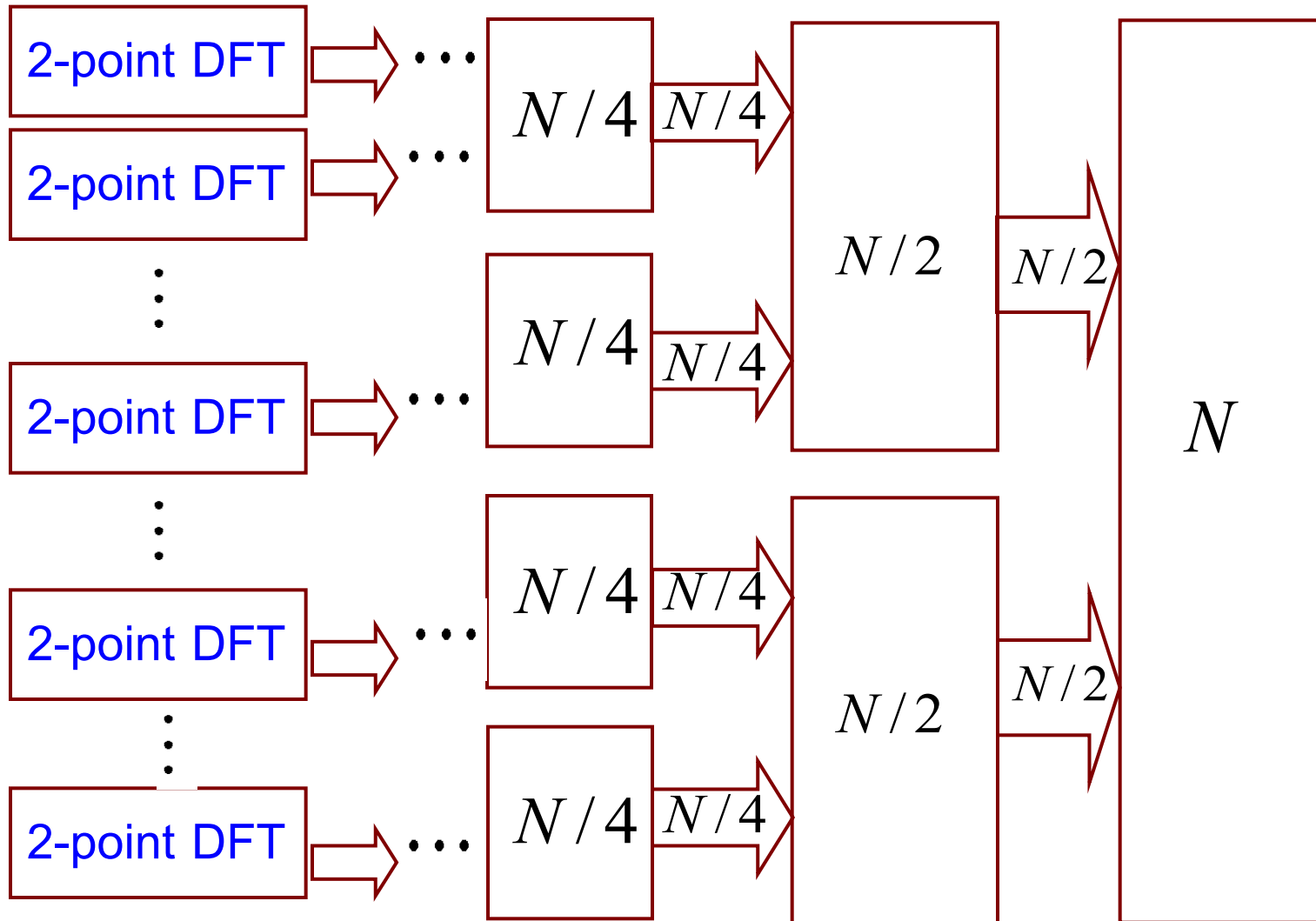
8-point DIT-FFT



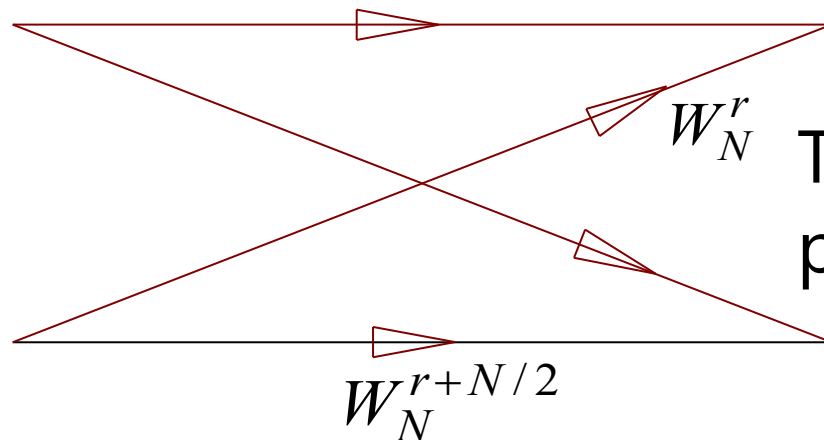
16-point DIT-FFT



N-point DIT-FFT

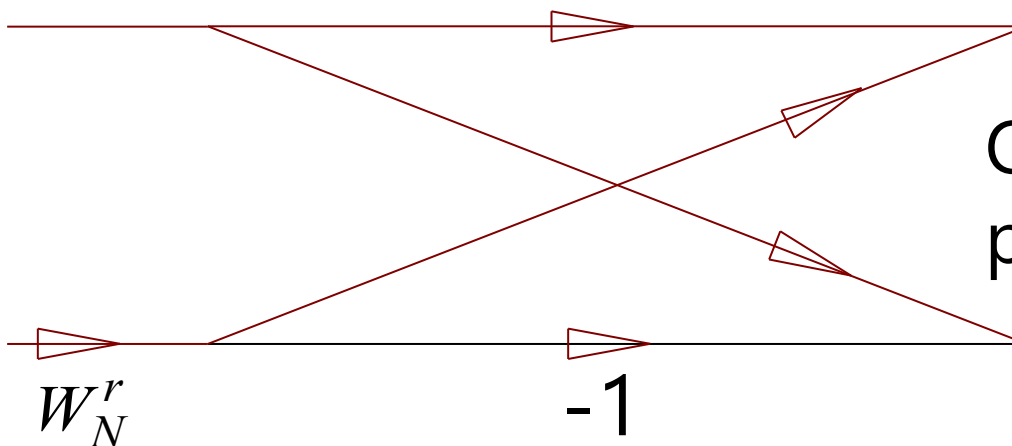


Modified Butterfly



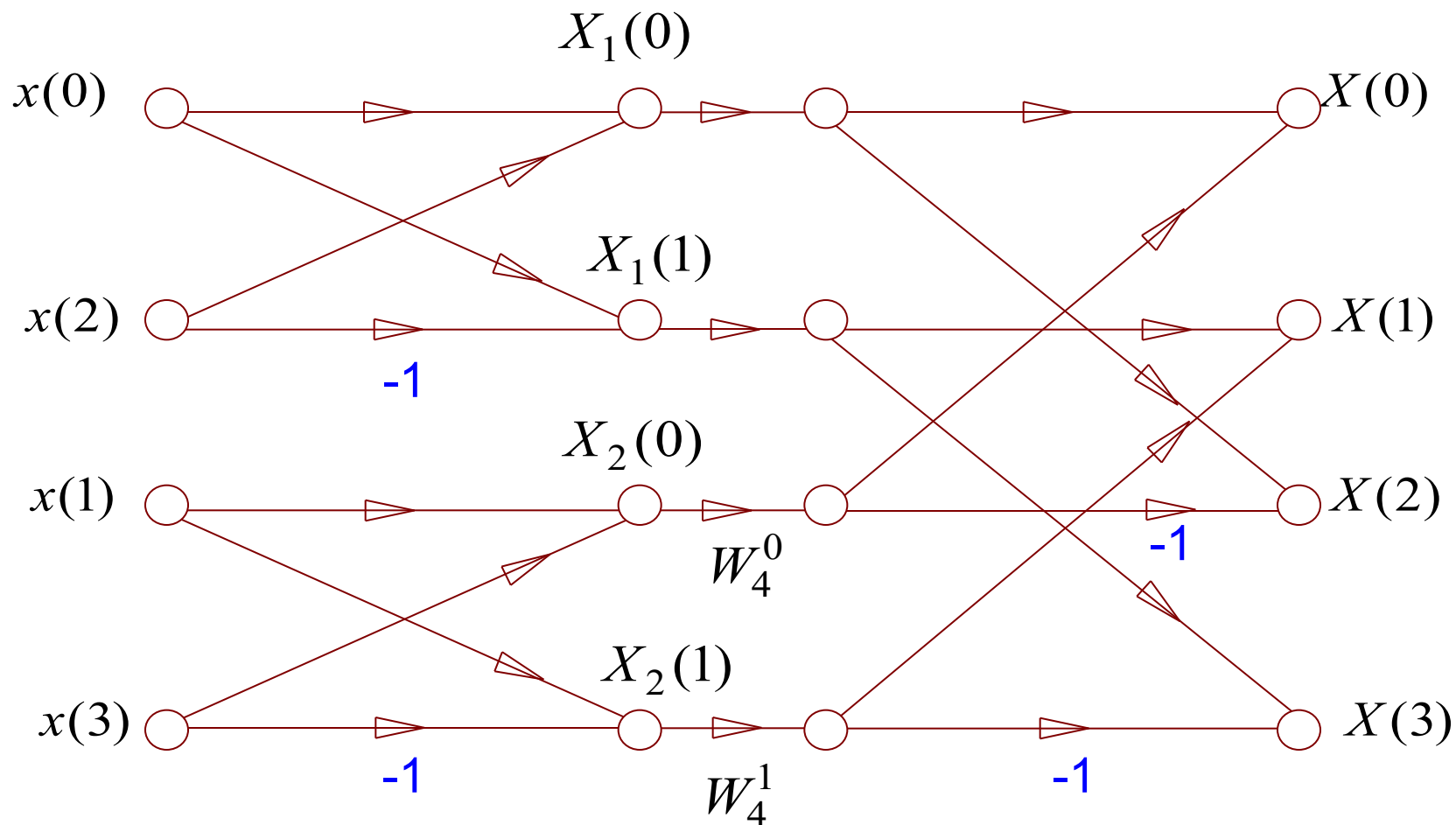
Two complex multiplies
per butterfly

$$W_N^{r+N/2} = W_N^r W_N^{N/2} = W_N^r (-1)$$



One complex multiply
per butterfly

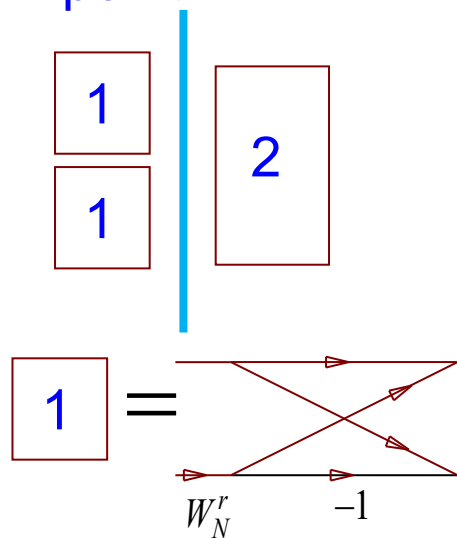
4-point DFT



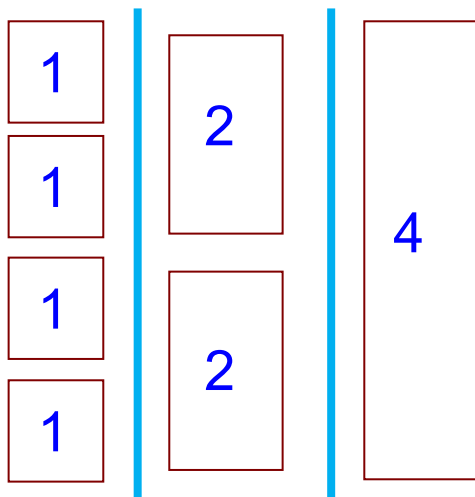
Number of Butterflies per Stage

16-point DFT = 32

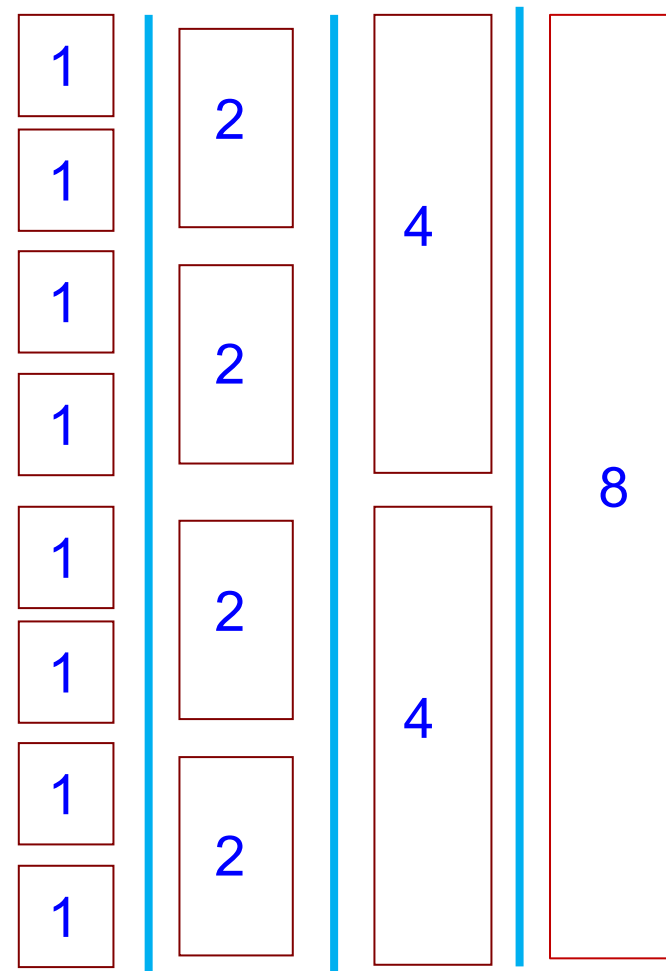
4-point DFT = 4



8-point DFT = 12



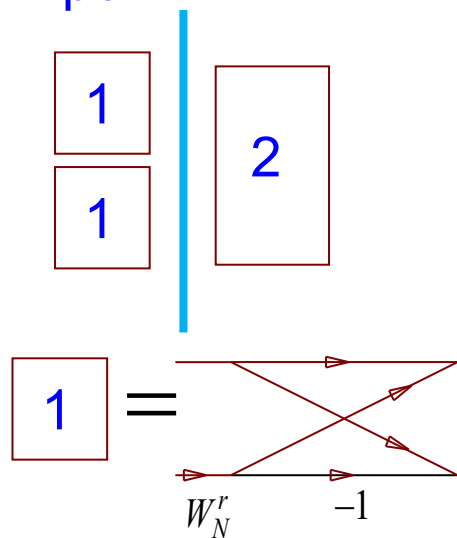
Number of Butterflies per stage
= $(N/2)$



Number of Stages (R)

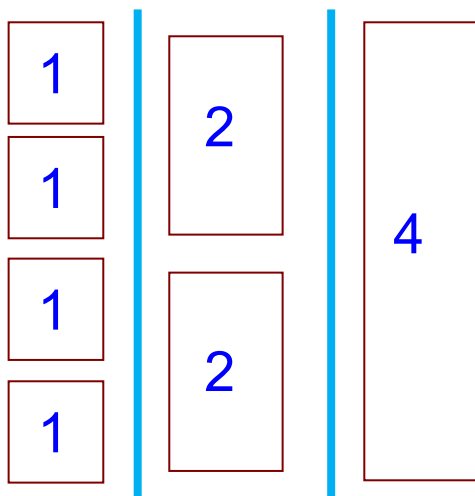
16-point DFT = 32

4-point DFT = 4

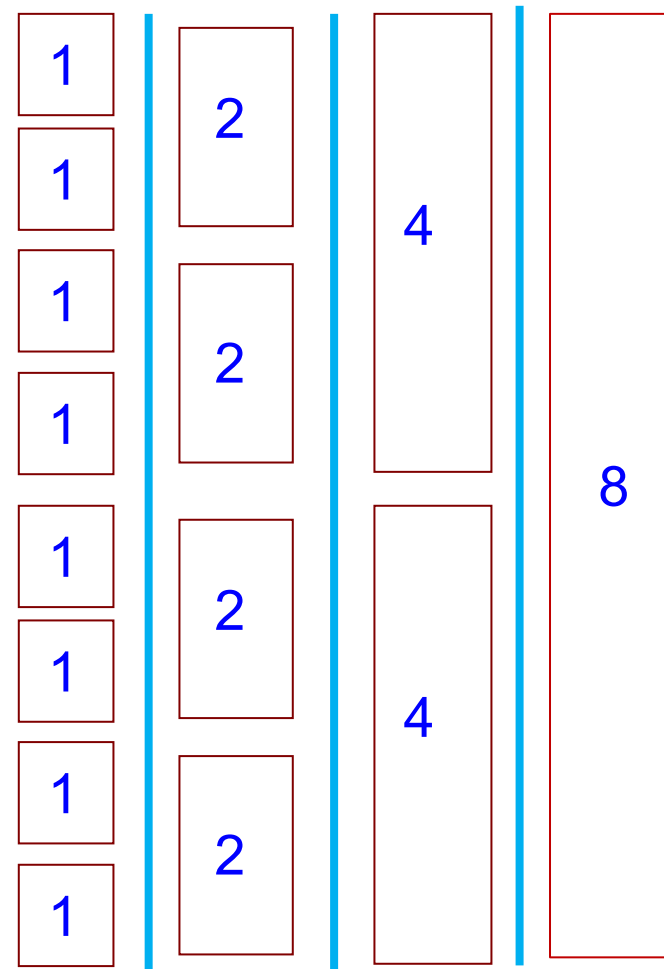


R= 2

8-point DFT = 12



R= 3



R= 4

Number of stages (R) = $\log_2 N$

Number of Complex Multiplies

- Number of Complex Multiplies = Number of Butterflies per stage ($N/2$) x Number of Stages x Number of Complex Multiplies per Butterflies

$$= \frac{N}{2} \times \log_2 N \times 1$$

$$= \frac{N}{2} \log_2 N$$

Number of Complex Multiplies

DFT vs. FFT

N	DFT N^2	FFT $(N/2 \log_2 N)$
2	4	1
4	16	4
8	64	12
:	:	:
256	65,536	1,024
512	262,144	2,304
1,024	1,048,576	5,120